EXPT:- 3a.
23-07-2026.

# UNDERSTANDING THE CISCO PACKET TRACER ENVIRONMENT.

AIM:-

To understand the Cisco Packet Tracer Environment and design a simple computer network using its interface and networking component.

STEPS:-

* Open the Cisco Packet Tracer application.
* Observe the user interface, including the menu bar, main toolbar, workspace, Common Tools Bar, Network Component Box, and Realtime/Simulation tabs.
* From the Network Component box, select the required network devices, such as PCs, Hub and Switch.
* Drag and drop the devices into the workspace.
* Connect the devices using copper straight through cables.
* Configure IP addresses for the PCs using the desktop → IP Configuration option.
* Verify That all links are active.
* Use the add simple PDU tool to send a packet between two PCs.
* Observe the packet transmission in realtime mode and simulation mode.
* Save the network topology.

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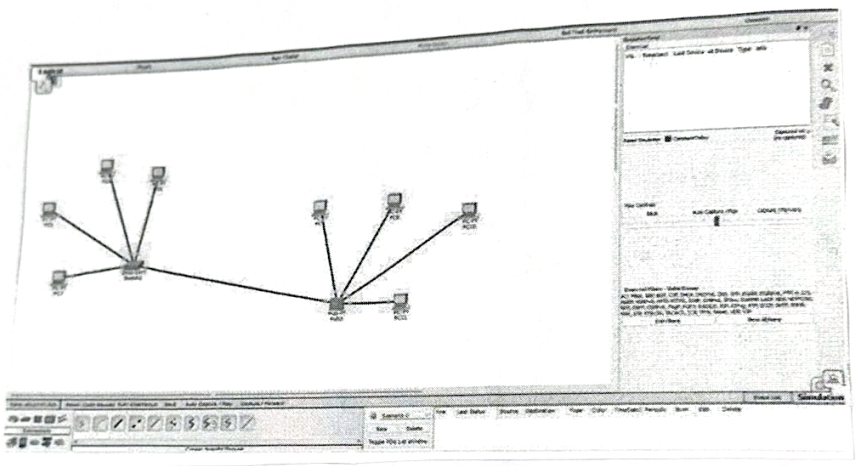


RESULT:-

Thus, The Cisco Packet Tracer environment was successfully explored and a simple network was designed. The interface components, device connections, IP configuration were successfully using both realtime and simulation modes.

EXPT:-3b1
23-07-2026.

# HUB.

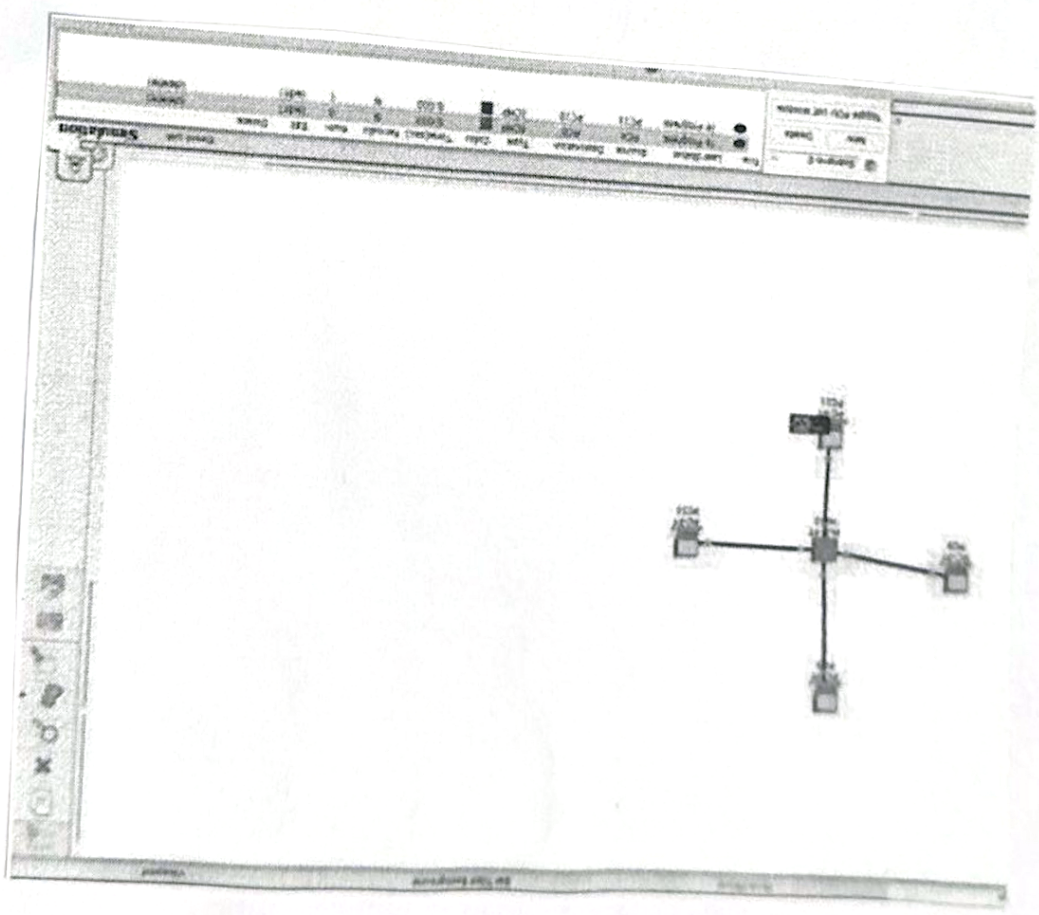
**AIM:-**
To study the behaviour of a HUB by observing how it forwards data packets using Cisco Packet Tracer.

**STEPS:-**

* Open Cisco Packet Tracer.
* Drag and place 1 Hub and 4 Generic PCs into the workspace.
* Connect all PCs to the Hub using Copper Straight-Through cables.
* Configure IP addresses for all PCs in the same network.
* Select add simple PDU from the common Tools Bar.
* Send a packet from one PC to the another.
* Switch to simulation mode and observe how the Hub forwards the packet.

**RESULT:-**

The Hub broadcasts the recieved packet to all connected devices. Every PC recieves the packet, but only the destination PC accepts it, while the other discard it.



EXPT:- 3b2
23-07-2026.

# SWITCH.

## AIM:-

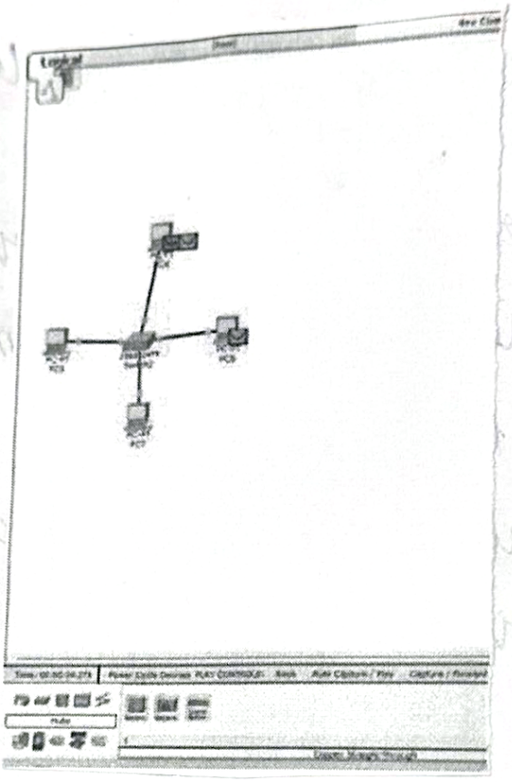
To study the behaviour of a switch by observing how it forwards data packets using Cisco Packet Tracer.

## STEPS:-

* Open Cisco Packet Tracer.
* Drag and place 1 Switch and 4 Generic PCs into the workspace.
* Connect all PCs to the switch using copper straight-through cables.
* Configure IP addresses for all the PCs in the same network.
* Select add simple PDU from the common tools bar.
* Send a packet from one PC to the another.
* Switch to simulation mode and observe how the switch forwards the packet.

## RESULT:-

The switch forwards the packet only to the destination PC by using its MAC address table. This reduces unnecessary network traffic and improves communication efficiency.



EXPT:-3b3.

# HUB AND SWITCH (COMPARISON).

23-07-2026.

AIM:-

To compare the packet forwarding behaviour of a Hub and switch using Cisco Packet Tracer

STEPS:-

* Create one network using a Hub and another using a switch, each connected to 4 PCs.
* Configure IP addresses for all PCs in the both networks
* Send a simple PDU between two PCs in same network.
* Observe packet transmission in simulation mode.
* Compare how the Hub and switch forward packets.

RESULT:-

The Hub broadcasts packets to all connected devices whereas the switch forwards packets only to the intended destination. Therefore, the switch provides the better performance, reduced network traffic and improved efficiency compared to the Hub.